MANAGEMENT INFORMATION SYSTEM

Database Design Document

Client-friendly explanation of how the MIS database is organized, what each section stores, and how the business data connects together.

Simple explanation: the database is like a set of connected registers. One register stores customers, another stores apartments, another stores sales, another stores employees, another stores warehouse stock, and so on. These registers are connected so the system can follow the full business process from customer management to sales, rentals, payroll, inventory, procurement, and approvals.

Main design goal: one system should support property operations, customer records, employee work, project usage, stock control, company assets, payments, and reporting without duplicating data.

1. Core Idea Of The Database

The system is divided into business sections. Each section contains tables that store one type of information. The most important sections are:

1. User access and security
2. Customers and apartments
3. Apartment sales and installment payments
4. Apartment rentals and rental payments
5. Employees and payroll
6. Projects, warehouses, materials, and company assets
7. Purchase requests, material requests, and asset requests
8. Documents, CRM messages, approvals, notifications, and offline sync support

2. Client-Friendly Module Design

A. User And Access Control

Table	Meaning
users	People who can log into the system
roles	User groups such as Admin, Sales, Accountant, Storekeeper
permissions	Actions that each role is allowed to perform
model_has_roles / model_has_permissions / role_has_permissions	Connection tables that assign roles and permissions
sessions / personal_access_tokens / password_reset_tokens	Login and token support tables

Logic: these tables control who can enter the system and what they are allowed to do.

B. Customers And Apartments

Table	Meaning
customers	People or organizations that buy or rent units
apartments	The apartment, shop, or unit register

Logic: a customer can buy an apartment or rent an apartment. Apartments are the main property units inside the system.

C. Apartment Sales

Table	Meaning
apartment_sales	Main sale contract
installments	Installment schedule of the sale
installment_payments	Actual installment payments received
apartment_sale_financials	Financial summary of the sale
municipality_payment_letters	Letter issued for municipality payment
municipality_receipts	Proof of municipality payments
apartment_sale_possession_logs	History of key handover or possession actions
apartment_sale_terminations	Sale cancellation/termination details

Logic: one sale belongs to one customer and one apartment. A sale can have many installments and many payments. The system also keeps the municipality share, financial balance, deed progress, and possession history.

D. Apartment Rentals

Table	Meaning
apartment_rentals	Rental agreement between tenant and apartment
rental_payments	Rent dues or billing rows
rental_payment_receipts	Actual rental payment receipts

Logic: a rental connects one tenant to one apartment for a contract period. The system tracks advance payment, monthly rent, payment status, and receipts.

E. Employees And Payroll

Table	Meaning
employees	Employee register
salary_advances	Salary taken in advance
salary_payments	Actual salary payment record

Logic: each employee can have many salary advances and many salary payments over time.

F. Projects, Warehouses, Materials, And Company Assets

Table	Meaning
projects	Project register
vendors	Supplier or vendor register
warehouses	Warehouse register
materials	Consumable goods such as cement, brick, paint, cable
warehouse_material_stocks	How much material is in each warehouse
project_material_stocks	How much material has already been issued to each project
stock_movements	Full history of materials moving in and out
company_assets	Reusable company-owned stock such as desktops, tools, machinery, vehicles

Logic: materials are consumable stock, while company assets are reusable stock. Warehouses hold both materials and available company assets.

> **Important current design note:** company assets are currently quantity-based stock lines. For example, one row can represent *Desktop stock = 100* rather than one separate row for each individual desktop.

G. Material, Purchase, And Asset Requests

Table	Meaning
material_requests	Request header for materials needed by a project/employee
material_request_items	Lines inside the material request
purchase_requests	Purchase request header for buying materials or assets
purchase_request_items	Lines inside the purchase request
asset_requests	Request header for reusable company assets
asset_assignments	Asset allocation and return history

Logic: employees and projects request materials or assets. If warehouse stock is not enough, the system creates a purchase request. When purchases are received, stock is added to the system. Assets are assigned and later returned or marked damaged/lost.

H. Documents, CRM, Notifications, Approvals, And Offline Support

Table	Meaning
documents	Uploaded files linked to customers, apartments, sales, or other modules
crm_messages	Customer communication history such as reminders
notifications	In-app notifications shown to users
approvals	Approval request header
approval_logs	Approval history
system_settings	System-wide settings
sync_inbox	Offline-first sync safety table

Logic: these tables support the rest of the business modules by handling files, communication, alerts, approvals, settings, and safe offline synchronization.

3. Simple Business Workflow

Sales Workflow

1. Create customer
2. Create apartment
3. Create apartment sale
4. Create installment schedule if needed
5. Record installment payments
6. Track municipality amount and receipts
7. Track deed progress and key handover
8. If sale is cancelled, keep the termination record

Rental Workflow

1. Create rental contract between customer and apartment
2. Generate rental payment rows
3. Record rental receipts when payments are received
4. Track key handover and contract ending

Material Workflow

1. Create material master record
2. Receive stock into a warehouse
3. Employee creates material request for a project
4. Approver approves the request
5. Warehouse stock is reserved
6. Storekeeper issues material to the project
7. Warehouse stock decreases and project stock increases
8. Movement is saved in stock history

Purchase Workflow

1. If warehouse stock is not enough, create purchase request
2. Purchase request can be for material or company asset
3. Approver approves the request
4. Goods are received into the warehouse
5. Material stock or company asset stock increases

Company Asset Workflow

1. Create company asset stock line

2. Employee or project requests asset
3. Approver approves request
4. Allocator assigns quantity from company asset stock
5. Assignment history is saved
6. On return, stock comes back or moves into damaged/lost quantities

4. Main Relationships In Plain Language

- One customer can have many apartment sales
- One customer can have many apartment rentals
- One apartment can appear in sales or rentals over time
- One sale can have many installments and many payment records
- One rental can have many payment rows and many receipts
- One warehouse can store many materials and many company assets
- One material can exist in many warehouses
- One project can use many materials
- One purchase request can contain many line items
- One asset request can create one or more asset assignment records
- One employee can create many requests and receive many payroll records

5. Simple ERD Diagram

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USERS
  ├─ manage access through ROLES and PERMISSIONS
  ├─ approve or process SALES, PURCHASES, ASSET REQUESTS, PAYMENTS
  └─ receive NOTIFICATIONS

CUSTOMERS
  ├─ buy through APARTMENT_SALES
  ├─ rent through APARTMENT_RENTALS
  ├─ receive CRM_MESSAGES
  └─ can have DOCUMENTS

APARTMENTS
  ├─ are sold through APARTMENT_SALES
  ├─ are rented through APARTMENT_RENTALS
  └─ can have DOCUMENTS

APARTMENT_SALES
  ├─ have INSTALLMENTS
  ├─ have INSTALLMENT_PAYMENTS
  ├─ have APARTMENT_SALE_FINANCIALS
  ├─ have MUNICIPALITY_PAYMENT_LETTERS
  ├─ have MUNICIPALITY_RECEIPTS
  ├─ have APARTMENT_SALE_POSSESSION_LOGS
  └─ may have APARTMENT_SALE_TERMINATIONS

APARTMENT_RENTALS
  ├─ have RENTAL_PAYMENTS
  └─ have RENTAL_PAYMENT_RECEIPTS

EMPLOYEES
  ├─ receive SALARY_ADVANCES
  ├─ receive SALARY_PAYMENTS
  ├─ create MATERIAL_REQUESTS
  ├─ create PURCHASE_REQUESTS
  └─ create ASSET_REQUESTS

WAREHOUSES
  ├─ hold WAREHOUSE_MATERIAL_STOCKS
  └─ hold COMPANY_ASSETS

MATERIALS
  ├─ exist in WAREHOUSE_MATERIAL_STOCKS
  ├─ move to PROJECT_MATERIAL_STOCKS
  ├─ appear in MATERIAL_REQUEST_ITEMS
  ├─ appear in PURCHASE_REQUEST_ITEMS
  └─ create STOCK_MOVEMENTS

COMPANY_ASSETS
  ├─ are requested in ASSET_REQUESTS
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  ├─ are assigned in ASSET_ASSIGNMENTS
  └─ may be increased by PURCHASE_REQUEST_ITEMS

PURCHASE_REQUESTS
  └─ contain PURCHASE_REQUEST_ITEMS for MATERIALS or COMPANY_ASSETS

PROJECTS
  ├─ use MATERIALS
  ├─ use COMPANY_ASSETS
  ├─ receive MATERIAL_REQUESTS
  └─ receive ASSET_REQUESTS
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6. Final Client Summary

The database is designed to support one complete management system for the business. It connects customers, apartments, sales, rentals, employees, payroll, projects, warehouses, materials, company assets, purchase requests, approvals, files, notifications, and offline sync. This allows the company to avoid duplicate data and follow every important business activity from start to finish.

Prepared from the implemented database structure in the current MIS backend codebase.